

Renal and Urinary disorders			Oliguria; dysuria; pollakiuria; urinary retention; polyuria; feeling of residual urine
General disorders and administration site conditions	Fever; rigors; malaise		Injection site induration

Like most cytotoxic agents bleomycin can give rise to both immediate and to delayed toxic effects. The most immediate effect is fever on the day of injection. Anorexia, tiredness or nausea also may occur. Pain at the injection site or in the region of the tumour has occasionally been reported, and other rare adverse effects are hypotension and local thrombophlebitis after intravenous administration.

Fever may develop with a lag time of 4-5 hours or more after the administration of this drug. Because a dose- response relation exists between the fever and dose at a given time, if the fever is severe, appropriate measures should be taken such as administering a reduced dose at shorter intervals, or antihistaminic and antipyretic agents before and/or after administration of this drug.

The majority of patients who receive a full course of bleomycin develop lesions of the skin or oral mucosa. Induration, hyperkeratosis, reddening, tenderness and swelling of the tips of the fingers, ridging of the nails, bulla formation over pressure points such as elbows, loss of hair and stomatitis are rarely serious and usually disappear soon after completion of the course.

The most serious delayed effect is interstitial pneumonia, which may develop during, or occasionally after, a course of treatment. This condition may sometimes develop into fatal pulmonary fibrosis, although such an occurrence is rare at recommended doses. Previous or concurrent radiotherapy to the chest is an important factor in increasing the incidence and severity of lung toxicity.

A few cases of acute fulminant reactions with hyperpyrexia and cardiorespiratory collapse have been observed after intravenous injections of doses higher than those recommended. Hypotension, hyperpyrexia and drug-related deaths have been reported rarely following intracavitary instillation of bleomycin.

During post-marketing surveillance the following events have been reported: sepsis, pancytopenia, thrombocytopenia, anaemia, neutropenia, chest pain, myocardial infarction, Raynaud's syndrome, embolism, thrombosis, digital ischaemia and cerebral infarction.

Symptoms and Treatment of overdose: The acute reaction to an over dosage of bleomycin would probably include hypotension, fever, rapid pulse and general symptoms of shock. Treatment is purely symptomatic. In the event of respiratory complications, the patient should be treated with a corticosteroid and a broad-spectrum antibiotic. There is no specific antidote to bleomycin.

Effects on Ability to Drive and Use Machine: This depends on the patient's condition and should be considered in cooperation with the doctor.

Instructions for Use

Incompatibilities: Bleomycin solution should not be mixed with solutions of essential amino acids, riboflavine, ascorbic acid, dexamethasone, aminophylline or frusemide.

Directions for reconstitution:

For intramuscular or subcutaneous injection: dissolve the contents of the vial in 1-5mL of 0.9% Sodium Chloride Injection or Sterile Water for Injection.

For intravenous or intra-arterial injection: dissolve the contents of the vial in 5-10mL of 0.9% Sodium Chloride Injection or Sterile Water for Injection and administer slowly over a period of 10 minutes.

Suitable diluents are Sterile Water for Injections, 0.9% Sodium Chloride Injection. Reconstituted solutions containing 150 to 15,000 IU (0.15 - 15units USP)/mL bleomycin prepared using the recommended diluents remain stable for periods of at least 24 hours when stored room temperatures (20°C - 25°C) and 10 days when stored at 2-8°C. However, in order to reduce the possibility of microbiological contamination, reconstituted injections should be used as soon as practicable after preparation. Any unused portions must be discarded in compliance with acceptable procedures for the disposal of anticancer drugs.

Storage Condition: Store at 2°C -8°C.

After Reconstitution: Reconstitute with 0.9% Sodium Chloride Injection or Sterile Water for Injection. The solution should be stable for up to 24 hours at 20-25°C and for 10 days at 2-8°C.

Shelf Life: 2 years

Therapeutical Code (If Any) Pharmacotherapeutic group:

Other cytotoxic antibiotics, ATC code: L01DC01

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To be sold by retail on the prescription of a cancer specialist / cancer hospital / institution only

BLEOSOL

Bleomycin Powder for Solution for Injection/Infusion 15 units

Name and Strength of Active Ingredient (s)

Bleomycin Sulfate USP equivalent to Bleomycin 15 Units

Dosage Form: Powder for Solution for Injection or Infusion (Lyophilised Powder)

Product Description:

Initial Description: White or cream-colored, lyophilized mass filled in colourless glass vials and sealed with rubber plugs and flip off aluminium seal.

BLEOSOL (Bleomycin Powder for Solution for Injection/ Infusion) is packed as 15 Units in 5 ml tubular clear glass vials (Type I), stoppered with 20 mm grey butyl rubber stoppers/plugs (Slotted) and 20 mm aluminium flip off seal. Each vial is packed into secondary carton along with a Package Insert.

Description after Reconstitution: A Clear colourless solution.

Diluent: No diluents are provided/ Supplied together with the product for reconstitution. 0.9% Sodium Chloride Injection or Sterile Water for Injection are used as diluents.

Pharmacodynamics:

ATC code: L01DC01, other cytotoxic antibiotics

Bleomycin is a basic, water-soluble glycopeptide with cytotoxic activity. The mechanism of action of bleomycin is believed to involve single-strand scission of DNA, leading to inhibition of cell division, of growth and of DNA synthesis in tumour cells.

Apart from its antibacterial and antitumour properties, bleomycin is relatively free from biological activity. When injected intravenously it may have a histamine-like effect on blood pressure and may cause a rise in body temperature.

Pharmacokinetics:

Bleomycin is administered parenterally. After intravenous (IV) administration of a bolus dose of 15 x 10³ IU/m² body surface, peak concentrations of 1 to 10 IU are achieved in plasma. Following the intramuscular (IM) injection of 15 x 10³ IU/m² peak plasma concentrations of about 1 IU/ml have been reported. The peak plasma concentration is reached 30 minutes after an IM injection. Continuous infusion of bleomycin 30 x 10³ IU/m² daily, for 4 to 5 days, resulted in an average steady state plasma concentration of 100-300 milli IU/mL. After IV injections of bleomycin in a dose of 15 x 10³ IU/m² body surface, the area under the serum concentration curve is, on average, 300 milli IU x min x ml⁻¹. Bleomycin is only bound to plasma proteins to a slight extent. Bleomycin is rapidly distributed in body tissues, with the highest concentrations in skin, lungs, peritoneum and lymph. Low concentrations are seen in the bone marrow. Bleomycin could not be detected in cerebrospinal fluid after intravenous injection. Bleomycin appears to cross the placental barrier.

The mechanism for bio-transformation is not yet fully known. Inactivation takes place during enzymatic breakdown by bleomycin hydrolase, primarily in plasma, liver and other organs and, to a much lesser degree, in skin and lungs. When bleomycin was administered as an IV bolus injection in a dose of 15 x 10³ IU/m² body surface, initial and terminal half-lives were 0.5 and 4 hours respectively.

Given as a continuous intravenous infusion in a dose of 30 x 10³ IU/m² daily for 4 to 5 days bleomycin disappears from plasma with initial and terminal half-lives of about 1.3 hours and 9 hours, respectively.

Given as a continuous intravenous infusion in a dose of 30 Units daily for 4 to 5 days bleomycin disappears from plasma with initial and terminal half-lives of about 1.3 hours and 9 hours, respectively. About two thirds of the administered drug is excreted unchanged in the urine, probably by glomerular filtration. Approximately 50% is recovered in the urine in the 24 hours following an IV or IM injection. The rate of excretion, therefore, is highly influenced by renal function; concentrations in plasma are greatly elevated if usual doses are given to patients with renal impairment with only up to 20% excreted in 24 hours. Observations indicate that it is difficult to eliminate bleomycin from the body by dialysis.

Indication:

Bleomycin is indicated for palliation and treatment adjuvant to surgery and radiation therapy of the following neoplasms:

- Squamous cell carcinoma of the skin, head and neck and oesophagus (primary indication)
- Squamous cell carcinoma of the larynx, penis and uterine cervix
- Squamous cell carcinoma of the bronchus (response infrequent)
- Choriocarcinoma and embryonal cell carcinoma of the testis
- Advanced Hodgkin's disease and other lymphomas
- Mycosis fungoides

Note: Use of bleomycin after radiation therapy is less successful than use before radiation therapy.

Bleomycin is bone marrow sparing and may be used when other cytotoxic agents are contraindicated.

Recommended Dose:

Routes of administration: Bleomycin may be given by the intramuscular, intravenous, subcutaneous or intra-arterial routes.

Note: Because of the possibility of an anaphylactoid reaction, lymphoma patients should receive test doses of between 1-5 units, for the first two treatments. If no acute allergic reaction occurs within 4-6 hours, the balance of the dose may be given. Thereafter the regular dosage schedule may be followed, if no reaction occurs.

Adult dose: Initial treatment (intramuscular, intravenous or subcutaneous administration) Total

doses of over 300,000 IU (300 units USP) should be given with great caution. 10,000 to 20,000 IU (10 - 20 units USP)/m² of body surface area given weekly or twice weekly. Alternatively, give 15,000 IU (15 units USP) daily for 7 days followed by three weeks off treatment and repeat twice so that a total dose of approximately 300,000 IU (300 units USP) is administered.

Improvement of lymphomas and testicular tumours is prompt ie. within two weeks while response by squamous cell cancers may take as long as three weeks. A therapeutic response should be observed as the total dose approaches 150,000 IU (150 units USP), if this is not seen, consideration should be given to other therapy.

Note: When bleomycin is used in combination with other antineoplastic agents, pulmonary toxicities may occur at lower doses.

Intra-arterial administration: Intra-arterial infusion/perfusion is employed when increased drug concentrations at the cancer site are desired. The suggested dosage schedule is 30,000 to 60,000 IU (30-60 units USP) once or twice a week until the total recommended dose of 300,000 IU (300 units USP) is reached.

Repeat treatment: In patients for whom a course of bleomycin treatment provides an initial but incomplete response, a repeat course is suggested. Patients who show superficial improvement after one course e.g. in cases of squamous cell carcinoma, may benefit from a second course of treatment to prevent recurrence.

A repeat course may be commenced after a minimum of 3-4 weeks following completion of the first course, providing no sign of pulmonary toxicity have been observed. A total dose of 150,000 IU (150 units USP) for repeat treatment is recommended.

Paediatric dose: No information available.

Geriatric dose: Adult dose should be used with caution, particularly in patients over 70 years.

In impaired liver function: Use adult dose with caution.

In impaired renal function: As bleomycin is mostly excreted unchanged and as there is a high correlation between renal bleomycin clearance and creatinine clearance, impairment of function may require reduction in dosage and careful monitoring for toxicity.

Dosage reductions of 40-75% have been recommended for patients with creatinine clearance values of 40 mL/min.

Directions for reconstitution:

For intramuscular or subcutaneous injection: dissolve the contents of the vial in 1-5mL of Sterile Water for Injection or 0.9% Sodium Chloride Injection.

For intravenous or intra arterial injection: dissolve the contents of the vial in 5-10mL of Sterile Water for Injection or 0.9% Sodium Chloride Injection and administer slowly over a period of 10 minutes.

Suitable diluents are Sterile Water for Injections, 0.9% Sodium Chloride Injection. Reconstituted solutions containing 150 to 15,000 IU (0.15 - 15unitsUSP)/mL bleomycin prepared using the recommended diluents remain stable for periods of at least 24 hours when stored room temperatures (20°C -25°C) and 10 days when stored at 2-8°C. However, in order to reduce the possibility of microbiological contamination, reconstituted injections should be used as soon as practicable after preparation. Any unused portions must be discarded in compliance with acceptable procedures for the disposal of anticancer drugs.

Route of Administration: Intramuscular, Intravenous, Subcutaneous, or Intra-Arterial

Contraindication:

Bleomycin is contra-indicated in patients with acute pulmonary infection or chest X-ray findings suggesting diffuse fibrotic changes or greatly reduced lung function.

Patients with a past history of hypersensitivity or idiosyncratic reaction to an analogue of bleomycin.

Warning and Precautions:

Pulmonary toxicity of bleomycin is both dose-related and age-related. It may also occur when lower doses are administered, especially in elderly patients, patients with reduced kidney function, preexisting lung disease, previous or concurrent radiotherapy to the chest and in patients who need administration of oxygen. It is significantly enhanced by thoracic radiation and by hyperoxia used during surgical anaesthesia.

The earliest symptom associated with pulmonary toxicity of bleomycin is dyspnoea. Fine rales are the earliest sign. If pulmonary changes are noted, treatment should be discontinued until it can be determined if they are drug related. Patients should be treated with broad spectrum antibiotics and corticosteroids.

Patients undergoing treatment with bleomycin should have chest X-rays weekly. These should continue to be taken for up to 4 weeks after completion of the course and patients should be kept under clinical review for approximately 2 months. If breathlessness or lung infiltrates appear, not obviously attributable to tumour or to co-existent lung disease, administration of the drug must be stopped immediately and patients should be treated with a corticosteroid and a broad spectrum antibiotic. High oxygen concentrations should be used with caution in these cases.

Lung function tests which use 100% oxygen should not be used in patients who have been treated with bleomycin. Lung function tests using less than 21% oxygen are recommended as an alternative.

When bleomycin has been administered pre-operatively, reduced oxygen concentrations should be used during operation and post operatively.

Patients should be carefully monitored under the following conditions and bleomycin dosage should be reduced or prolong the dose interval based on clinical observation of the patient: These clinical conditions include the following:

- Patients treated previously or concurrently with radiation to the chest may develop more frequent or severe toxicity.
- Use with caution in patients with significant renal impairment as clearance may be reduced and toxicity increased.
- Use with caution in patients with severe heart disease or hepatic dysfunction as toxicity may be increased.
- Use with caution in patients with varicella as fatal systematic dysfunctions may occur.

Because bleomycin treatment may give rise to shock, if any abnormalities appear, withdraw bleomycin immediately, and take appropriate measures. (Because shock is likely to develop in patients with malignant lymphomas at the 1st – 2nd administration, you may start this drug treatment with lower dose and after establishing that no acute reactions to the drug occur,

increase the dose to the usual level).

With long-term administration of bleomycin, peplomycin or other analogies of bleomycin, toxicity is thought to be additive, thus administration must be performed with care.

Attention should be paid to the appearance or exacerbation of infection and any bleeding tendency.

In adults or adolescents capable of reproduction, effects on the sexual glands should be considered.

Intravenous administration: Vascular pain may occur, therefore, it is important to pay due attention to concentration of the injection and administration rate. Give intravenously as slowly as possible.

Intramuscular administration: Avoid repeated injections at the same site and innervated sites, particularly if administering to paediatrics. If insertion of the injection needle evokes intense pain or if blood flows back into the syringe, withdraw the needle immediately and inject at a different site.

Interaction with Other Medicaments:

When bleomycin is used as one of the drugs in multiple chemotherapy regimes the toxicity of bleomycin should be borne in mind in the selection and dosage of drugs with similar toxic potential.

The addition of other cytotoxic drugs can necessitate changes and dose alterations. Increased pulmonary toxicity has been noted when bleomycin is given with cisplatin.

Previous or concurrent radiotherapy to the chest and/or administration of anti-tumour agents (e.g. cisplatin) are important factors in increasing the incidence and severity of lung toxicity such as interstitial pneumonia or pulmonary fibrosis.

Previous or concurrent radiotherapy to the head or neck is a factor increasing stomatitis and angular stomatitis may deteriorate. It may cause inflammation of pharyngolaryngeal mucosa infrequently resulting in hoarseness

Because of bleomycin's sensitisation of lung tissue, patients who have received bleomycin preoperatively are at greater risk of developing pulmonary toxicity when oxygen is administered at surgery and a reduction in inspired oxygen concentration during operation and post-operatively is recommended.

In patients treated for testicular cancer with a combination of bleomycin and vinca alkaloids a syndrome has been reported corresponding to Raynaud's disease, ischaemia which can lead to necrosis of peripheral parts of the body (fingers, toes, nose tip).

The following clinical incompatibilities have been noted:-Cytotoxics possibly reduce the absorption of phenytoin. Concomitant use of bleomycin with clozapine should be avoided due to an increased risk of agranulocytosis.

Pregnancy and Lactation

Pregnancy: The administration of this drug to pregnant patients, or women suspected of being pregnant, is not recommended. The use of bleomycin should be avoided whenever possible during pregnancy, particularly during the first trimester.

Breastfeeding: Bleomycin should not be given to mothers who are breast feeding.

Fertility: Bleomycin can cause congenital malformations. Women should not become pregnant during and six months after treatment.

Side Effects:

The most frequently observed adverse reactions in 1613 patients receiving bleomycin were pulmonary manifestations such as interstitial pneumonia or pulmonary fibrosis (10.2%), sclerosis of skin, pigmentation (40.6%), fever and rigors (39.8%), alopecia (29.5%), anorexia and weight decrease (28.7%), general malaise (16.0%), nausea and vomiting (14.6%), stomatitis (13.3%) and nail changes (11.2%).

System Organ Class	Very Common >1/10	Common > 1/100 to <1/10	Uncommon > 1/1,000 to < 1/100
Neoplasms, Benign, Malignant and Unspecified (including Cysts and Polyps)			Pain at the site of tumour
Blood and lymphatic system disorders			Leukopenia
Metabolism and Nutrition disorders	Anorexia		
Nervous system disorders		Headache	Dizziness
Vascular disorders		Haemorrhage	Shock; vein wall hypertrophy; venous stenosis
Respiratory, thoracic and mediastinal disorders	Interstitial pneumonia; pulmonary fibrosis		
Gastrointestinal disorders	Weight decrease; nausea; vomiting; stomatitis	Angular stomatitis	Diarrhoea
Hepatobiliary disorders			Hepatocellular injury
Skin and subcutaneous tissue	Alopecia; skin hypertrophy; pigmentation; deformation and discolouration of the nail; scratch dermatitis	Rash; urticaria; erythroderma	
Musculoskeletal and connective tissue disorders	Scleroderma		